Balancing Redox Equations 1

Essential Pre-Uni Chemistry K3.1

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Balance the following redox equations.

Use the lowest possible **integer** coefficients your answers.

Part A

(a)

$$\mathrm{H_2\,(g) + ?Ag^+\,(aq) \longrightarrow ?H^+\,(aq) + ?Ag\,(s)}$$

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

Click in the box below to edit your answer.

Part B

(b)

$$\mathrm{?Fe\,(s) + ?Cl_2\,(g) \longrightarrow ?FeCl_3\,(s)}$$

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

Click in the box below to edit your answer.

Part C

(c)

$$I_2\,(aq) + ?S_2O_3{}^{2-}\,(aq) \longrightarrow ?I^-\,(aq) + S_4O_6{}^{2-}\,(aq)$$

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

Click in the box below to edit your answer.

Part D

(d)

$$CH_4\,(g) + ?Cl_2\,(g) \longrightarrow CCl_4\,(g) + ?HCl\,(g)$$

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

Click in the box below to edit your answer.

Part E

(e)

$$C_6H_{12}O_6\,(aq) + ?O_2\,(g) \longrightarrow ?CO_2\,(g) + ?H_2O\,(l)$$

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

Click in the box below to edit your answer.

Part F

(f)

$$?\mathrm{K}\,(\mathrm{s}) + ?\mathrm{H_2O}\,(\mathrm{l}) \longrightarrow ?\mathrm{KOH}\,(\mathrm{aq}) + \mathrm{H_2}\,(\mathrm{g})$$

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

Click in the box below to edit your answer.

Part G

(g)

$$?\mathrm{Co^{3+}}\,(\mathrm{aq}) + ?\mathrm{H_2O}\,(\mathrm{l}) \longrightarrow ?\mathrm{Co^{2+}}\,(\mathrm{aq}) + ?\mathrm{H^+}\,(\mathrm{aq}) + \mathrm{O_2}\,(\mathrm{g})$$

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

Click in the box below to edit your answer.

Part H

(h)

$$a\,\mathrm{Fe^{2+}}\,(\mathrm{aq}) + b\,\mathrm{MnO_4{}^-}\,(\mathrm{aq}) + c\,\mathrm{H^+}\,(\mathrm{aq}) \longrightarrow d\,\mathrm{Fe^{3+}}\,(\mathrm{aq}) + e\,\mathrm{Mn^{2+}}\,(\mathrm{aq}) + f\,\mathrm{H_2O}\,(\mathrm{l})$$

This chemical equation is too long to fit on your screen, so please balance the equation and give your answer as a string of numbers in the order of $abcdef$. For example, if the question is

$$a\,\mathrm{H_2}\,(\mathrm{g}) + b\,\mathrm{Ag^+}\,(\mathrm{aq}) \longrightarrow c\,\mathrm{H^+}\,(\mathrm{aq}) + d\,\mathrm{Ag}\,(\mathrm{s}),$$

and you think the answer is

$$3\,\mathrm{H_2}\,(\mathrm{g}) + 1\,\mathrm{Ag^+}\,(\mathrm{aq}) \longrightarrow 6\,\mathrm{H^+}\,(\mathrm{aq}) + 5\,\mathrm{Ag}\,(\mathrm{s}),$$

then input 3165.

Part I

(i)

$$a\,\mathrm{Zn}\,(\mathrm{s}) + b\,\mathrm{VO_2}^{+}\,(\mathrm{aq}) + c\,\mathrm{H^+}\,(\mathrm{aq}) \longrightarrow d\,\mathrm{Zn^{2+}}\,(\mathrm{aq}) + e\,\mathrm{V^{2+}}\,(\mathrm{aq}) + f\,\mathrm{H_2O}\,(\mathrm{l})$$

This chemical equation is too long to fit on your screen, so please balance the equation and give your answer as a string of numbers in the order of $abcdef$. For example, if the question is

$$a\,\mathrm{H_2}\,(\mathrm{g}) + b\,\mathrm{Ag^+}\,(\mathrm{aq}) \longrightarrow c\,\mathrm{H^+}\,(\mathrm{aq}) + d\,\mathrm{Ag}\,(\mathrm{s}),$$

and you think the answer is

$$3\,\mathrm{H_2}\,(\mathrm{g}) + 1\,\mathrm{Ag^+}\,(\mathrm{aq}) \longrightarrow 6\,\mathrm{H^+}\,(\mathrm{aq}) + 5\,\mathrm{Ag}\,(\mathrm{s}),$$

then input 3165.

Part J

(j)

$$\mathrm{C_2O_4H_2}\,(\mathrm{aq}) + \mathrm{H_2O_2}\,(\mathrm{aq}) \longrightarrow \mathbf{?}\mathrm{CO_2}\,(\mathrm{g}) + \mathbf{?}\mathrm{H_2O}\,(\mathrm{l})$$

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

Click in the box below to edit your answer.

Part K

(k)

$$a\,\mathrm{BaFeO_4}\,(\mathrm{s}) + b\,\mathrm{HCl}\,(\mathrm{aq}) \longrightarrow c\,\mathrm{BaCl_2}\,(\mathrm{aq}) + d\,\mathrm{FeCl_3}\,(\mathrm{aq}) + e\,\mathrm{H_2O}\,(\mathrm{l}) + f\,\mathrm{Cl_2}\,(\mathrm{g})$$

This chemical equation is too long to fit on your screen, so please balance the equation and give your answer as a string of numbers in the order of $abcdef$. For example, if the question is

$$a\,\mathrm{H_2}\,(\mathrm{g}) + b\,\mathrm{Ag^+}\,(\mathrm{aq}) \longrightarrow c\,\mathrm{H^+}\,(\mathrm{aq}) + d\,\mathrm{Ag}\,(\mathrm{s}),$$

and you think the answer is

$$3\,\mathrm{H_2}\,(\mathrm{g}) + 1\,\mathrm{Ag^+}\,(\mathrm{aq}) \longrightarrow 6\,\mathrm{H^+}\,(\mathrm{aq}) + 5\,\mathrm{Ag}\,(\mathrm{s}),$$

then input 3165.

Part L

(l)

$$a\,\mathrm{CH_3CH_2CH_2OH}\,(\mathrm{l}) + b\,\mathrm{Cr_2O_7}^{2-}\,(\mathrm{aq}) + c\,\mathrm{H^+}(\mathrm{aq}) \longrightarrow d\,\mathrm{CH_3CH_2CHO}\,(\mathrm{l}) + e\,\mathrm{Cr^{3+}}\,(\mathrm{aq}) + f\,\mathrm{H_2O}\,(\mathrm{l})$$

This chemical equation is too long to fit on your screen, so please balance the equation and give your answer as a string of numbers in the order of $abcdef$. For example, if the question is

$$a\,\mathrm{H_2}\,(\mathrm{g}) + b\,\mathrm{Ag^+}\,(\mathrm{aq}) \longrightarrow c\,\mathrm{H^+}\,(\mathrm{aq}) + d\,\mathrm{Ag}\,(\mathrm{s}),$$

and you think the answer is

$$3\,\mathrm{H_2}\,(\mathrm{g}) + 1\,\mathrm{Ag^+}\,(\mathrm{aq}) \longrightarrow 6\,\mathrm{H^+}\,(\mathrm{aq}) + 5\,\mathrm{Ag}\,(\mathrm{s}),$$

then input 3165.

Question deck:

STEM SMART Chemistry Week 11

Balancing Redox Equations 2

Essential Pre-Uni Chemistry K3.2

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Complete the balanced equations to show the reactions between the following pairs of substances in acidic aqueous conditions (no fractions).

Part A

Manganate(VII) and hydrogen peroxide

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

$$5\,H_2O_2\,(aq) + 2\,MnO_4{}^-\,(aq) + \ ? \longrightarrow 2\,Mn^{2+}\,(aq) + 8\,H_2O\,(l) + \ ?$$

Click in the box below to edit your answer.

Part B

Scandium and ethanoic acid

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

$$2\,Sc\,(s) + \ ?CH_3COOH\,(aq) \longrightarrow \ ? + 3\,H_2\,(g)$$

Click in the box below to edit your answer.

Part C

Ethanol and boiling dichromate(VI)

$$3\,CH_3CH_2OH\,(aq) + 2\,Cr_2O_7{}^{2-}\,(aq) + a\,H^+\,(aq) \longrightarrow 3\,CH_3COOH\,(aq) + b\,Cr^{3+}\,(aq) + c\,H_2O\,(l)$$

This chemical equation is too long to fit on your screen, so please balance the equation and give your answer as a string of numbers in the order of abc. For example, if the question is

$$a\,H_2\,(g) + b\,Ag^+\,(aq) \longrightarrow 6\,H^+\,(aq) + c\,Ag\,(s),$$

and you think the answer is

$$3\,H_2\,(g) + 1\,Ag^+\,(aq) \longrightarrow 6\,H^+\,(aq) + 5\,Ag\,(s),$$

then input 315.

Part D

Chlorate(V) and chloride

Please **click on and drag** the pre-loaded species to create a balanced chemical equation.

$$ClO_3{}^-\,(aq) + \ ?Cl^-\,(aq) + \ ?H^+\,(aq) \longrightarrow \ ?Cl_2\,(g) + \ ?$$

Click in the box below to edit your answer.

Question deck:

STEM SMART Chemistry Week 11

Balancing Redox Equations 3

Essential Pre-Uni Chemistry K3.3

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P2

Complete the balanced equations to show the reactions between the following pairs of substances in alkaline aqueous conditions (no fractions).

Part A

Sulfite and bromate(V)

sulfite and bromate(V)

$$3\,SO_3^{\,2-}(aq) + \ ?BrO_3^{\,-}(aq) \longrightarrow 3\,SO_4^{\,2-}(aq) + \ ?$$

Click in the box below to edit your answer.

Part B

Hydrogen peroxide and chromium(III)

hydrogen peroxide and chromium(III)

$$3\,H_2O_2\,(aq) \ + \ ?Cr^{3+}(aq) \ + \ ?OH^-(aq) \longrightarrow \ ?CrO_4^{\,2-}(aq) \ + \ ?$$

Click in the box below to edit your answer.

Part C

Chlorate(I) and iron(III)

chlorate(I) and iron(III)

$$?ClO^{-}\,(aq) \;+\; ?Fe(OH)_3\,(s) \longrightarrow\; ?FeO_4{}^{2-}\,(aq) \;+\; ?Cl_2\,(g) + \;?\; + \;?$$

Click in the box below to edit your answer.

Part D

Manganate(VI) and methanoate

manganate(VI) and methanoate

$$?HCOO^{-}\,(aq) + MnO_4{}^{2-}\,(aq) \longrightarrow MnO_2\,(s) + OH^{-}\,(aq) + \;?$$

Click in the box below to edit your answer.

Question deck:

STEM SMART Chemistry Week 11

Oxidation of Thiosulfate

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C1

An aqueous solution contains $1\,\mathrm{mol}$ of $S_2O_3^{2-}$ ions and this reduces $4\,\mathrm{mol}$ of Cl_2 molecules to Cl^- ions. What is the sulfur-containing product of this reaction?

- S
- SO_2
- SO_3^{2-}
- SO_4^{2-}
- $S_4O_6^{2-}$

Adapted with permission from UCLES, A-Level Chemistry, June 1989, Paper 3, Question 20

Question deck:
STEM SMART Chemistry Week 11

Disproportionation 1

Essential Pre-Uni Chemistry K4.1

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

The following are all examples of disproportionation reactions.

Part A

(a)

$$Cl_2\,(aq) + H_2O\,(l) \longrightarrow HCl\,(aq) + HOCl\,(aq)$$

State the symbol of the element that is disproportionated in the reaction.

State the element's oxidation state when it first appears in the chemical equation.

State the element's oxidation state the second time it appears in the chemical equation.

State the element's oxidation state the third time it appears in the chemical equation.

Part B

(b)

$$Cu_2SO_4\,(aq) \longrightarrow Cu\,(s) + CuSO_4\,(aq)$$

State the symbol of the element that is disproportionated in the reaction.

State the element's oxidation state when it first appears in the chemical equation.

State the element's oxidation state the second time it appears in the chemical equation.

State the element's oxidation state the third time it appears in the chemical equation.

Part C

(c)

$$2\,CO\,(g) \longrightarrow C\,(s) + CO_2\,(g)$$

State the symbol of the element that is disproportionated in the reaction.

State the element's oxidation state when it first appears in the chemical equation.

State the element's oxidation state the second time it appears in the chemical equation.

State the element's oxidation state the third time it appears in the chemical equation.

Part D

(d)

$$2\,\mathrm{HOF}\,(\mathrm{aq}) \longrightarrow \mathrm{H_2O}\,(\mathrm{l}) + \mathrm{OF_2}\,(\mathrm{g})$$

State the symbol of the element that is disproportionated in the reaction.

State the element's oxidation state when it first appears in the chemical equation.

State the element's oxidation state the second time it appears in the chemical equation.

State the element's oxidation state the third time it appears in the chemical equation.

Part E

(e)

$$5\,MnO_4{}^{2-}\,(aq) + 8\,H^+\,(aq) \longrightarrow Mn^{2+}\,(aq) + 4\,MnO_4{}^-\,(aq) + 4\,H_2O\,(l)$$

State the symbol of the element that is disproportionated in the reaction.

State the element's oxidation state when it first appears in the chemical equation.

State the element's oxidation state the second time it appears in the chemical equation.

State the element's oxidation state the third time it appears in the chemical equation.

Question deck:

STEM SMART Chemistry Week 11

Disproportionation 2

Essential Pre-Uni Chemistry K4.2

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Complete and balance the following equations that represent disproportionation reactions.

Part A

(a)

$$?H_2O_2\,(aq) \longrightarrow ?H_2O\,(l) + _____$$

Click in the box below to edit your answer.

Part B

(b)

$$?I_2\,(aq) + ?OH^-\,(aq) \longrightarrow _____ + 5\,I^-\,(aq) + 3\,H_2O\,(l)$$

Please **click on and drag** the pre-loaded species in the equation editor to create your chemical equation.

Click in the box below to edit your answer.

Part C

(c)

$$8\,S_2O_3{}^{2-}\,(aq) + 16\,H^+\,(aq) \longrightarrow S_8\,(s) + \underline{\hspace{3cm}} + 8\,H_2O\,(l)$$

Please **click on and drag** the pre-loaded species in the equation editor to create your chemical equation.

Click in the box below to edit your answer.

Question deck:

STEM SMART Chemistry Week 11

Iron in a Nail

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

A nail of mass $1.40\,\mathrm{g}$ was dissolved in an excess of dilute sulfuric acid to form $100\,\mathrm{cm^3}$ of solution. A $10\,\mathrm{cm^3}$ sample of this solution required $4.0 \times 10^{-4}\,\mathrm{mol}$ of manganate (VII) for complete oxidation.

In acidic solution:

$$\mathrm{MnO_4}^- + 8\,\mathrm{H^+} + 5\,\mathrm{e^-} \longrightarrow \mathrm{Mn^{2+}} + 4\,\mathrm{H_2O}$$

By assuming that, in dissolving in sulfuric acid, the iron in the nail was converted entirely into $\mathrm{Fe^{2+}}\,(\mathrm{aq})$ and that manganate (VII) oxidises $\mathrm{Fe^{2+}}$ to $\mathrm{Fe^{3+}}$, calculate:

Part A

Moles of $\mathrm{Fe^{2+}}$

The number of moles of $\mathrm{Fe^{2+}}$ produced from the nail.

Part B

% of Fe

The percentage of iron in the nail.

Adapted with permission from UCLES, A-Level Chemistry, June 1992, Paper 3, Question 2

Question deck:

STEM SMART Chemistry Week 11

Sodium Nitrite

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C1

Nitrogen can be obtained in the laboratory by warming a mixture of ammonium chloride and sodium nitrite, $NaNO_2$. Water is also produced and a solid is left.

Part A

Ammonium chloride and sodium nitrite

Suggest an identity for the solid.

Part B

Equation

Write a balanced equation for the reaction, including state symbols, balancing to obtain the lowest integer coefficients possible.

Part C

Ammonium chloride and sodium nitrate

A similar reaction takes place when ammonium chloride is heated with sodium nitrate $NaNO_3$ but this time the only different product is an oxide of nitrogen. Suggest a formula for this oxide, and the oxidation state of nitrogen in it.

Formula:

Oxidation state:

Adapted with permission from UCLES, A-Level Chemistry, June 1994, Paper 1, Question 7

Question deck:

STEM SMART Chemistry Week 11

High Oxidation States

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C1

The maximum possible oxidation state of an element could occur if all the outermost electrons, the so-called valence electrons, were used in bonding. The maximum number of valence electrons is equal to the group number for Groups 1 to 11, and the group number minus 10 for elements from Groups 12 to 18. Note the maximum possible oxidation state is not always achievable; however, each of the following elements forms an oxide exhibiting the theoretical maximum oxidation state for that element. Give the formula for each oxide.

Part A

Xenon (Xe)

Part B

Polonium (Po)

Part C

Chlorine (Cl)

Part D

Niobium (Nb)

Part E

Osmium (Os)

Part F

Yttrium (Y)

Adapted with permission from the Cambridge Chemistry Challenge 2021, Question 2

Question deck:

STEM SMART Chemistry Week 11